

Jinwoo Chung

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RESEARCH INTERESTS

Efficient and Trustworthy Foundation Models, Quantization (PTQ/QAT), Model Compression, VLMs, Diffusion Models, KV-Cache Quantization, Low-Bit Deployment.

EDUCATION

Kookmin University, Korea

Aug. 2025 – Feb. 2027 (expected)

M.S. in *Computer Engineering* (GPA: 4.31/4.5)

- Adviser: Prof. Jangho Kim

Kookmin University, Korea

Mar. 2022 – Aug. 2025

B.S. in *Artificial Intelligence* (GPA: 4.0/4.5)

CURRENT RESEARCH

Behavior-Aware Quantization for Foundation Models

2026 – Present

- Studying how low-bit quantization changes generative behavior in diffusion models, especially minority generation and sample diversity.
- Analyzing how activation quantization alters attention patterns and cross-modal routing in vision-language models.
- Exploring behavior-preserving quantization methods for efficient and reliable foundation model deployment.

PUBLICATIONS

<International Conferences>

1. **Jinwoo Chung**, Sangho An, Sungyeop Jung, Jangho Kim, "Temporal-Aware Quantization via Sequence-Wise Quantizer Bounds Ensembling for Real-World Video Super-Resolution.", The 19th European Conference on Computer Vision (ECCV 2026) *Acceptance rate: 27.5%.*
2. **Jinwoo Chung**, Sangho An, Hyunsung Park, Jangho Kim, "SR-BEVS: Sparse Recovery Bird's-Eye View Model Soup for Robust Camera-Based 3D Object Detection.", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026) *Acceptance rate: 36%.*
3. **Jinwoo Chung**, Sungyeop Jung, Weronika Czorapinska, Jangho Kim, "Quantized Model Soup Shake-Up: Weight Perturbation for Enhanced Ensemble Diversity", ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026) *Acceptance rate: 18.5%.*
4. HyeonJun Lee*, Hyeonsik Jo*, **Jinwoo Chung**, Jangho Kim, "SQuaT: Self-Supervised Knowledge Distillation via Student-Aware Quantized Teacher Features", The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026) *Acceptance rate: 28%.*

<Under Review>

1. “Hardware-Friendly Token-Group Activation Quantization for Low-Bit Mamba Super-Resolution”
Jinwoo Chung, Jangho Kim. (*NeurIPS 2026 Under review.*)
2. “Gradient Interaction Sensitivity for Structured Pruning of Large Language Models”
Sangho An*, **Jinwoo Chung***, Jangho Kim. (*NeurIPS 2026 Under review.*)
3. “Quantized Audio Spectrogram Transformer Soup: Robustness and Deployment Efficiency.”
Jinwoo Chung, Songha Lee, Jangho Kim. (Short paper under review)

HONORS & AWARDS

Outstanding Paper Award, ICROS 2026

2026

Awarded for “Reinforcement Learning-Based Quantization Scale Optimization for Low-Bit Vision Transformers”

PROJECTS

On-Device AI Quantization Algorithm Development

Jan. 2025 – Dec. 2025

Industry-Academia Project (LG Electronics)

- Developed quantization algorithms for on-device AI deployment.
- Submitted a research paper proposing three quantization methods.

Autonomous Driving Perception Robustness

Jan. 2025 – Dec. 2025

Industry-Academia Project (Hyundai Motor Company)

- Built a road-icing segmentation dataset for low-temperature driving scenarios.
- Developed a robustness-enhancing method for camera-based BEV 3D object detection.